

07 Slowly Varying Weights for Rolling Validation

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Research question and choice of weights

Analysis 06 studied power weights $w_i = i^\alpha$. Here I ask whether $\log i$, a polynomial of $\log i$, $\log \log i$, and slowly decreasing weights lead to consistent selection in the same Gaussian example.

Candidate weight functions

The seven main weights cover a baseline, slowly increasing weights, and two decreasing weights.

Weight	Why test it?	Expected role
1	Unweighted baseline	Reference
$\log(i+1)$	Check the log idea	Slow growth
$[\log(i+1)]^2$	Higher log power	Stronger slow growth
$1 + \log(i+1) + [\log(i+1)]^2$	Check a polynomial of log	Genuine polynomial of log
$\log \log(i+e)$	Check the log-log idea	Extremely slow growth
$1/\log(i+1)$	Test decreasing weights	Possible boundary case
$1/[\log(i+1)]^2$	Test a faster decrease	Compare the ratio criterion

Why use shifted logarithms?

The shifts keep every weight defined and positive at small indices: $1/\log 1$ is undefined, while $i+e > e$ implies $\log \log(i+e) > 0$. They do not change the asymptotics because $\log(i+1) \sim \log i$ and $\log \log(i+e) \sim \log \log i$.

The family $w_i = [\log(i+1)]^p$ contains five of the main cases. I also examine $p = -1.25, -0.75, -0.5, 0.5$ near the boundary and include $i^{0.01}$ as a reference to Analysis 06.

Common setup and decision criterion

What counts as a wrong selection?

Let $Y_i = \theta_\star + \xi_i$, where the ξ_i are independent $N(0, 1)$, and put $S_m = \sum_{t=1}^m \xi_t$. At time i , compare $\hat{\theta}_{i-1} = (i-1)^{-1} \sum_{t < i} Y_t$ with the oracle $\theta_\star$. For positive deterministic weights w_i , their weighted squared-loss difference is

$$\Xi_n = \sum_{i=2}^n w_i \left\{ \frac{S_{i-1}^2}{(i-1)^2} - 2\xi_i \frac{S_{i-1}}{i-1} \right\}. \quad (1)$$

A positive score favors the oracle; $\Xi_n < 0$ is a wrong selection.

What the Chebyshev ratio can establish

Write

$$M_n = \mathbb{E}\Xi_n, \quad V_n = \text{Var}(\Xi_n), \quad R_n = \frac{V_n}{M_n^2}.$$

Since $M_n > 0$, Chebyshev's inequality gives

$$\mathbb{P}(\Xi_n < 0) \leq \min\{1, R_n\}. \quad (2)$$

Thus $R_n \rightarrow 0$ proves consistency. A positive ratio limit is inconclusive; it does not prove inconsistency.

One formula for all candidate weights

The following formula applies to every deterministic weight considered here.

Proposition 1. For any positive deterministic weights, define

$$\begin{aligned} M_n &= \sum_{i=2}^n \frac{w_i}{i-1}, \\ a_{n,t} &= \sum_{i=t+1}^n \frac{w_i}{(i-1)^2}, \quad 1 \leq t < n, \\ b_{n,l} &= \sum_{i=l+1}^n \frac{w_i}{(i-1)^2} - \frac{w_l}{l-1}, \quad 2 \leq l \leq n. \end{aligned}$$

Then

$$\begin{aligned}\Xi_n - M_n &= \sum_{t=1}^{n-1} a_{n,t}(\xi_t^2 - 1) + 2 \sum_{l=2}^n b_{n,l} \sum_{k < l} \xi_k \xi_l, \\ V_n &= 2 \sum_{t=1}^{n-1} a_{n,t}^2 + 4 \sum_{l=2}^n (l-1) b_{n,l}^2.\end{aligned}\tag{3}$$

Proof. The sample mean at time i uses only earlier observations, so S_{i-1} is independent of ξ_i . This gives $\mathbb{E}(\xi_i S_{i-1}) = 0$ and $\mathbb{E}S_{i-1}^2 = i-1$, yielding the mean. To obtain the variance, expand S_{i-1}^2 in (1) and gather the terms containing each ξ_t^2 and each $\xi_k \xi_l$. The centered squares have variance 2, and each product $\xi_k \xi_l$ has variance 1. Distinct centered squares are uncorrelated. A square and a product are uncorrelated because some independent centered Gaussian variable appears to an odd power. The same reasoning applies to two different products, including products sharing one index. There are $l-1$ products with larger index l , so summing their variances gives (3). $\square$

For the weights considered here, it helps to separate coefficients that would remain if validation continued indefinitely from the change caused by stopping at n . Suppose $\sum_{i=2}^{\infty} w_i/(i-1)^2 < \infty$, and define

$$A_t = \sum_{i=t+1}^{\infty} \frac{w_i}{(i-1)^2}, \quad B_l = A_l - \frac{w_l}{l-1}, \quad T_n = A_n.$$

Here A_t is the infinite future tail of a diagonal coefficient, $T_n = A_n$ is the tail removed by stopping at n , and B_l is the off-diagonal residual after subtracting the direct validation contribution. The quantity B_l is important because its leading terms cancel.

Then $a_{n,t} = A_t - T_n$ and $b_{n,l} = B_l - T_n$ exactly. Since the weights are positive, $0 \leq a_{n,t} \leq A_t$. Using $(x-y)^2 \leq 2x^2 + 2y^2$ in (3) gives

$$V_n \leq 2 \sum_{t < n} A_t^2 + 8 \sum_{l=2}^n (l-1) B_l^2 + 4n(n-1) T_n^2.\tag{4}$$

The first sum is the diagonal contribution; the other terms bound the off-diagonal contribution.

For the log-power weights, the diagonal part stays bounded.

If the sum of $\sum_{t < n} A_t^2$, $\sum_{l=2}^n (l-1) B_l^2$, and $n^2 T_n^2$ is $o(M_n^2)$, then $R_n \rightarrow 0$ and selection is consistent.

Key point. This condition does not require increasing weights; it only requires the mean signal to dominate the variance. This is why decreasing weights are included.

What happens in the log-power family?

Consider $w_i = [\log(i+1)]^p$, with fixed $p \in \mathbb{R}$, and write $L = \log n$.

Proposition 2. For the log-power family, the ratio tends to zero exactly when $p \geq -1$. For decreasing weights, I use two limiting constants: $M_\infty(p) = \sum_{i=2}^\infty [\log(i+1)]^p / (i-1)$ is used only for $p < -1$, and, for $p < 0$,

$$V_\infty(p) = 2 \sum_{t=1}^\infty A_t^2 + 4 \sum_{l=2}^\infty (l-1) B_l^2 \in (0, \infty).$$

The first is a finite mean for $p < -1$. The second is the finite variance reached when $p < 0$.

Roadmap

The mean and variance each have three regimes:

$$M_n \begin{cases} \sim L^{p+1}/(p+1), & p > -1, \\ \sim \log L, & p = -1, \\ \rightarrow M_\infty(p), & p < -1, \end{cases} \quad V_n \begin{cases} \sim 2L^{2p}, & p > 0, \\ \rightarrow \pi^2, & p = 0, \\ \rightarrow V_\infty(p), & p < 0. \end{cases}$$

The mean has three regimes. The variance has three regimes. Because their break-points differ, the ratio has five cases: $p > 0$, $p = 0$, $-1 < p < 0$, $p = -1$, and $p < -1$.

Case 1: $p > 0$ – increasing log powers

The tested log and squared-log weights correspond to $p = 1$ and $p = 2$. In this case,

$$M_n \sim \frac{L^{p+1}}{p+1}, \quad V_n \sim 2L^{2p}, \quad R_n \sim \frac{2(p+1)^2}{L^2} \rightarrow 0.$$

The ratio vanishes, so selection is consistent.

Case 2: $p = 0$ – the unweighted reference

Here $w_i = 1$, $M_n \sim L$, and $V_n = 6 \sum_{m=1}^{n-1} m^{-2} \rightarrow \pi^2$. Consequently, $R_n \sim \pi^2/L^2 \rightarrow 0$. The mean diverges while the variance stays bounded, so the baseline is consistent.

Case 3: $-1 < p < 0$ – slowly decreasing weights

For example, $p = -1/2$ gives $w_i = 1/\sqrt{\log(i+1)}$. These weights decrease to zero, yet

$$M_n \sim \frac{L^{p+1}}{p+1} \rightarrow \infty, \quad V_n \rightarrow V_\infty(p), \quad R_n \sim \frac{V_\infty(p)(p+1)^2}{L^{2p+2}} \rightarrow 0.$$

The mean diverges and the variance stays bounded. **Decreasing weights can still be consistent.**

Case 4: $p = -1$ – the boundary case

For $w_i = 1/\log(i+1)$,

$$M_n \sim \log L = \log \log n, \quad V_n \rightarrow V_\infty(-1), \quad R_n \sim \frac{V_\infty(-1)}{(\log L)^2} \rightarrow 0.$$

The mean diverges only as $\log \log n$, so the ratio vanishes slowly. This boundary case is still consistent.

Case 5: $p < -1$ – a positive ratio limit

The main comparison $p = -2$ gives $w_i = 1/[\log(i+1)]^2$. Now $M_n \rightarrow M_\infty(p) > 0$ and $V_n \rightarrow V_\infty(p) > 0$, so $R_n \rightarrow V_\infty(p)/M_\infty(p)^2 > 0$. The total mean signal stops growing and the variance also has a finite positive limit. **A positive ratio limit does not prove inconsistency.**

The boundary in one statement

$$\boxed{p = -1 \text{ is the Chebyshev-ratio boundary for } [\log(i+1)]^p.} \quad (5)$$

Specifically, $p \geq -1 \Rightarrow R_n \rightarrow 0$, while $p < -1 \Rightarrow R_n \rightarrow$ a positive constant. On the first side, (2) proves consistency. On the second, the ratio provides no conclusion about the limiting wrong-selection probability.

Why $p = -1$? For $p \geq -1$ the mean still diverges; for $p < -1$ it converges. Meanwhile, the variance is bounded for $p < 0$, and for $p > 0$ it grows more slowly than M_n^2 . This places the ratio boundary at $p = -1$. As in Analysis 06, a positive ratio limit only makes the Chebyshev argument inconclusive.

Why the five cases have these limits

Proof. The mean is the easiest part: each validation time contributes $w_i/(i-1)$. Replacing $[\log(i+1)]^p/(i-1)$ by $(\log i)^p/i$ changes the partial sum by a bounded amount: the termwise error is $O((\log i)^p + (\log i)^{p-1}/i^2)$, which is summable. The integral $\int (\log x)^p dx/x$, with $u = \log x$, then shows exactly when those contributions add up without bound:

$$\begin{aligned} M_n &\sim L^{p+1}/(p+1), & p > -1, \\ M_n &\sim \log L, & p = -1, \\ M_n &\rightarrow M_\infty(p) \in (0, \infty), & p < -1. \end{aligned} \tag{6}$$

For the variance, the leading terms cancel:

$$A_l \sim \frac{(\log l)^p}{l}, \quad \frac{w_l}{l-1} \sim \frac{(\log l)^p}{l}, \quad B_l = A_l - \frac{w_l}{l-1}.$$

Thus B_l is smaller than either term separately. More precisely,

$$\begin{aligned} A_t &\sim \frac{(\log t)^p}{t}, & T_n &\sim \frac{L^p}{n}, \\ |B_l| &\leq C_p \left\{ \frac{(\log l)^{p-1}}{l} + \frac{(\log l)^p}{l^2} \right\}. \end{aligned} \tag{7}$$

Here and below constants may depend on fixed p . To estimate A_t , compare its tail sum with $I_p(t) = \int_t^\infty (\log x)^p x^{-2} dx$. On each unit interval the derivative bound gives a total comparison error $O((\log t)^p/t^2)$. Integration by parts gives $I_p(t) = (\log t)^p/t + pI_{p-1}(t)$. Also $I_q(t) = O((\log t)^q/t)$ for every fixed q : substitute $x = te^v$, so the remaining integral is $t^{-1} \int_0^\infty (\log t + v)^q e^{-v} dv$. The leading term is $(\log t)^p/t$, proving the first line of (7).

To see the cancellation in B_l , replace $(i-1)^{-2}$ in A_l by $[i(i-1)]^{-1}$; the error is $O((\log l)^p/l^2)$. Summation by parts gives

$$\sum_{i=l+1}^\infty \frac{w_i}{i(i-1)} = \frac{w_{l+1}}{l} + \sum_{i=l+2}^\infty \frac{w_i - w_{i-1}}{i-1}.$$

The difference between w_{l+1}/l and $w_l/(l-1)$ is $O((\log l)^p/l^2 + (\log l)^{p-1}/l^2)$. The mean-value theorem bounds $|w_i - w_{i-1}| \leq C_p (\log i)^{p-1}/i$. Applying the preceding integral bound to the remaining sum proves (7). **The leading term in B_l cancels.** For $p = 0$ this derivative term is zero and $B_l = O(l^{-2})$.

Now suppose $p > 0$. The diagonal part is bounded because $a_{n,t} \leq A_t$ and $\sum_t A_t^2 < \infty$ by (7). For the off-diagonal part, define

$$D_n = \sum_{l=2}^n (l-1) B_l^2.$$

The quantity D_n collects the B_l^2 contribution. It lets us compare that contribution with the terminal term involving T_n^2 .

From (7), the first term in the bound for B_l determines the larger order:

$$(l-1)B_l^2 = O\left(\frac{(\log l)^{2p-2}}{l}\right), \quad D_n = O\left(\sum_{l=2}^n \frac{(\log l)^{2p-2}}{l}\right).$$

Integral comparison reduces the sum to

$$\int^n \frac{(\log x)^{2p-2}}{x} dx = \int^{\log n} u^{2p-2} du, \quad u = \log x.$$

The integral is bounded, logarithmic, or polynomial according as $2p-2$ is less than, equal to, or greater than -1 . Solving $2p-2 = -1$ gives the threshold $p = 1/2$. Hence

$$D_n = \begin{cases} O(1), & 0 < p < 1/2, \\ O(\log \log n), & p = 1/2, \\ O(L^{2p-1}), & p > 1/2. \end{cases}$$

The three-way split is only needed for the common conclusion

$$\boxed{D_n = o(L^{2p})}.$$

Next, $T_n \sim L^p/n$ and $\sum_{l=2}^n (l-1) \sim n^2/2$, so

$$T_n^2 \sum_{l=2}^n (l-1) \sim \frac{L^{2p}}{n^2} \frac{n^2}{2} = \frac{L^{2p}}{2}.$$

After the factor 4 in (3), the terminal contribution is

$$4T_n^2 \sum_{l=2}^n (l-1) \sim 2L^{2p}.$$

Thus the terminal term has order L^{2p} , larger than D_n .

Since

$$(B_l - T_n)^2 = B_l^2 - 2B_l T_n + T_n^2,$$

the cross term satisfies

$$2 \left| T_n \sum_{l=2}^n (l-1) B_l \right| \leq 2 \left\{ \left(\sum_{l=2}^n (l-1) B_l^2 \right) \left(T_n^2 \sum_{l=2}^n (l-1) \right) \right\}^{1/2}.$$

Equivalently, the right side is $2\{D_n T_n^2 \sum_{l=2}^n (l-1)\}^{1/2}$. Since $D_n = o(L^{2p})$ and $T_n^2 \sum (l-1) \asymp L^{2p}$, the cross term is $o(L^{2p})$. Thus Cauchy–Schwarz makes it asymptotically smaller than the terminal term.

The four variance contributions now have orders

- diagonal part: $O(1)$;
- D_n part: $o(L^{2p})$;
- cross term: $o(L^{2p})$;
- terminal part, including the factor 4: $\sim 2L^{2p}$.

Therefore,

$$\boxed{V_n \sim 2L^{2p}, \quad p > 0.}$$

For $p < 0$, $\sum (l-1) B_l^2$ converges, whereas $n^2 T_n^2 \sim L^{2p} \rightarrow 0$. The same cross-term bound tends to zero. Dominated convergence for the diagonal sum, and convergence of the B_l series for the off-diagonal sum, give $V_n \rightarrow V_\infty(p)$. This limit is strictly positive because its first term includes $2A_1^2 > 0$.

At $p = 0$, the unit weights give $V_n = 6 \sum_{m=1}^{n-1} m^{-2} \rightarrow \pi^2$. For computation, if $W_{n-1} = \sum_{i=2}^{n-1} w_i$, then

$$V_n - V_{n-1} = \frac{2w_n}{(n-1)^2} \{(2n-1)w_n - 2W_{n-1}\}. \quad (8)$$

This recurrence also follows from the quadratic-form matrix: its old block gains $w_n \mathbf{1}\mathbf{1}^\top / (n-1)^2$, its new off-diagonal entries are $-w_n / (n-1)$, and the sum of all entries of the old matrix is $-W_{n-1}$. For unit weights (8) is $6/(n-1)^2$, starting from $V_1 = 0$. Combining these variance limits with (6) proves (5). $\square$

Hence

$$V_n \begin{cases} \sim 2L^{2p}, & p > 0, \\ \rightarrow \pi^2, & p = 0, \\ \rightarrow V_\infty(p), & p < 0. \end{cases}$$

The variance has three regimes: the terminal term dominates for $p > 0$, the unit-weight case has a special identity, and the variance stays bounded for $p < 0$.

The case $p = 0$ has its own leading constant. For every fixed $p > 0$, the terminal contribution eventually leads; with unit weights it does not. The positive- p constant therefore cannot simply be evaluated at $p = 0$.

Polynomial and log-log weights

Polynomial of log

The tested polynomial is

$$w_i = 1 + \log(i+1) + [\log(i+1)]^2.$$

More generally, for $w_i = P(\log(i+1))$ with $P(x) = cx^k + \text{lower-order terms}$, the highest-degree term dominates.

Proposition 3. Suppose $w_i = P(\log(i+1)) > 0$, where P is a polynomial of degree $k \geq 1$ with positive leading coefficient c . Then

$$M_n \sim \frac{c}{k+1} L^{k+1}, \quad V_n \sim 2c^2 L^{2k}, \quad R_n \sim \frac{2(k+1)^2}{L^2}. \quad (9)$$

Thus a polynomial of log reduces asymptotically to the log-power case $p = k$. For the tested degree-two polynomial, $R_n \sim 18/L^2 \rightarrow 0$.

Proof. The leading term controls the weight because $P(x) = cx^k + O(x^{k-1})$, so $P(\log(i+1)) = c[\log(i+1)]^k + O([\log(i+1)]^{k-1})$. When the terms are integrated to obtain the mean, lower degrees contribute only $O(L^k)$, compared with the leading $cL^{k+1}/(k+1)$. They also make a smaller contribution to the tail coefficient A_t : $O((\log t)^{k-1}/t)$, lower order than $c(\log t)^k/t$. The derivative of the weight is $O((\log x)^{k-1}/x)$, so the proof of (7) gives $A_t \sim c(\log t)^k/t$ and $B_l = O((\log l)^{k-1}/l + (\log l)^k/l^2)$. Thus $T_n = A_n \sim cL^k/n$, and $\sum_{l=2}^n (l-1) \sim n^2/2$ gives

$$4T_n^2 \sum_{l=2}^n (l-1) \sim 4 \frac{c^2 L^{2k}}{n^2} \frac{n^2}{2} = 2c^2 L^{2k}.$$

The lower-degree terms contribute only lower orders to both moments. The diagonal term is bounded, while $\sum_{l=2}^n (l-1)B_l^2 = O(L^{2k-1}) = o(L^{2k})$. The cross term is also lower order by the Cauchy–Schwarz estimate in Proposition 2. Equation (3) therefore gives (9), with the highest degree determining the leading orders. $\square$

Log-log weight

The exact weight $w_i = \log \log(i+e)$ grows more slowly than $\log i$.

Proposition 3 (continued). For this weight,

$$M_n \sim L \log L, \quad V_n \sim 2(\log L)^2, \quad R_n \sim \frac{2}{L^2}. \quad (10)$$

The common $(\log L)^2$ factor cancels from the ratio.

Proof. The mean differs by a bounded amount from $\int^n \log \log x \, dx/x$. With $u = \log x$ and $du = dx/x$, $\int \log u \, du = u \log u - u$. Hence $M_n = L \log L - L + O(1) \sim L \log L - L \sim L \log L$, where $L = \log n$. The derivative of the shifted weight is $1/((x+e) \log(x+e))$. The same tail comparison and summation by parts give

$$A_t \sim \frac{\log \log t}{t}, \quad B_l = O\left(\frac{1}{l \log l} + \frac{\log \log l}{l^2}\right).$$

Thus $T_n = A_n \sim \log \log n/n = (\log L)/n$, so

$$4T_n^2 \sum_{l=2}^n (l-1) \sim 4 \frac{(\log L)^2}{n^2} \frac{n^2}{2} = 2(\log L)^2.$$

The sums $\sum A_t^2$ and $\sum (l-1)B_l^2$ stay bounded. The same Cauchy–Schwarz argument used above makes the cross term smaller than the terminal term. Equation (3) therefore gives $V_n \sim 2(\log L)^2$, proving (10). $\square$

Dividing by the squared mean gives

$$R_n \sim \frac{2(\log L)^2}{L^2(\log L)^2} = \frac{2}{L^2} = \frac{2}{(\log n)^2} \rightarrow 0.$$

Thus the log-log weight is consistent.

Theory summary for the seven tested weights

Table 1 puts the seven main weights side by side. Here $L = \log n$, and each finite positive variance limit depends on the particular weight. The last column records what the Chebyshev ratio establishes.

Table 1: Theory summary for the seven main weights; $L = \log n$.

Weight	Mean behavior	Variance behavior	Ratio behavior	Conclusion
1	$\sim L$	$\rightarrow \pi^2$	$\sim \pi^2/L^2$	Consistent
$\log(i+1)$	$\sim L^2/2$	$\sim 2L^2$	$\sim 8/L^2$	Consistent
$[\log(i+1)]^2$	$\sim L^3/3$	$\sim 2L^4$	$\sim 18/L^2$	Consistent
$1 + \log(i+1) + [\log(i+1)]^2$	$\sim L^3/3$	$\sim 2L^4$	$\sim 18/L^2$	Consistent
$\log \log(i+e)$	$\sim L \log L$	$\sim 2(\log L)^2$	$\sim 2/L^2$	Consistent
$1/\log(i+1)$	$\sim \log L$	$\rightarrow V_\infty(-1) > 0$	$\asymp (\log L)^{-2}$	Consistent
$1/[\log(i+1)]^2$	$\rightarrow M_\infty(-2) > 0$	$\rightarrow V_\infty(-2) > 0$	$\rightarrow V_\infty(-2)/M_\infty(-2)^2 > 0$	Not decided

Numerical verification

Simulation design and checks

The simulations use centered Gaussian observations and the weights listed above. Exact moments extend to larger n ; simulation checks finite-sample agreement.

There are 5,000 independent replications, seed 2026, processed in batches of 250. The sample sizes are 100, 200, 500, 1000, 2000, 5000, 10000, 20000, 50000, 100000. Each Gaussian path is reused across weights and sample sizes, so the comparisons are dependent. Exact trajectories use 169 sample sizes, approximately log-spaced from 50 through 10^7 , including the simulation sizes. Equation (8) makes this calculation linear in the largest sample size.

The implementation of (3) was checked against a quadratic-form matrix at $n = 2, 3, 7, 19$ for every weight, including the mean against the matrix trace, the variance against twice the squared Frobenius norm, the score against the existing increment function, and (8) against (3). The largest absolute discrepancy is $5.7\text{e-}14$. The exact unweighted variance and seeded reproducibility were also checked.

Do the empirical variances agree with the exact ratios?

Blue curves are exact ratios; orange points use simulated variances divided by the exact squared mean. The horizontal axis is logarithmic and panels have separate vertical scales.

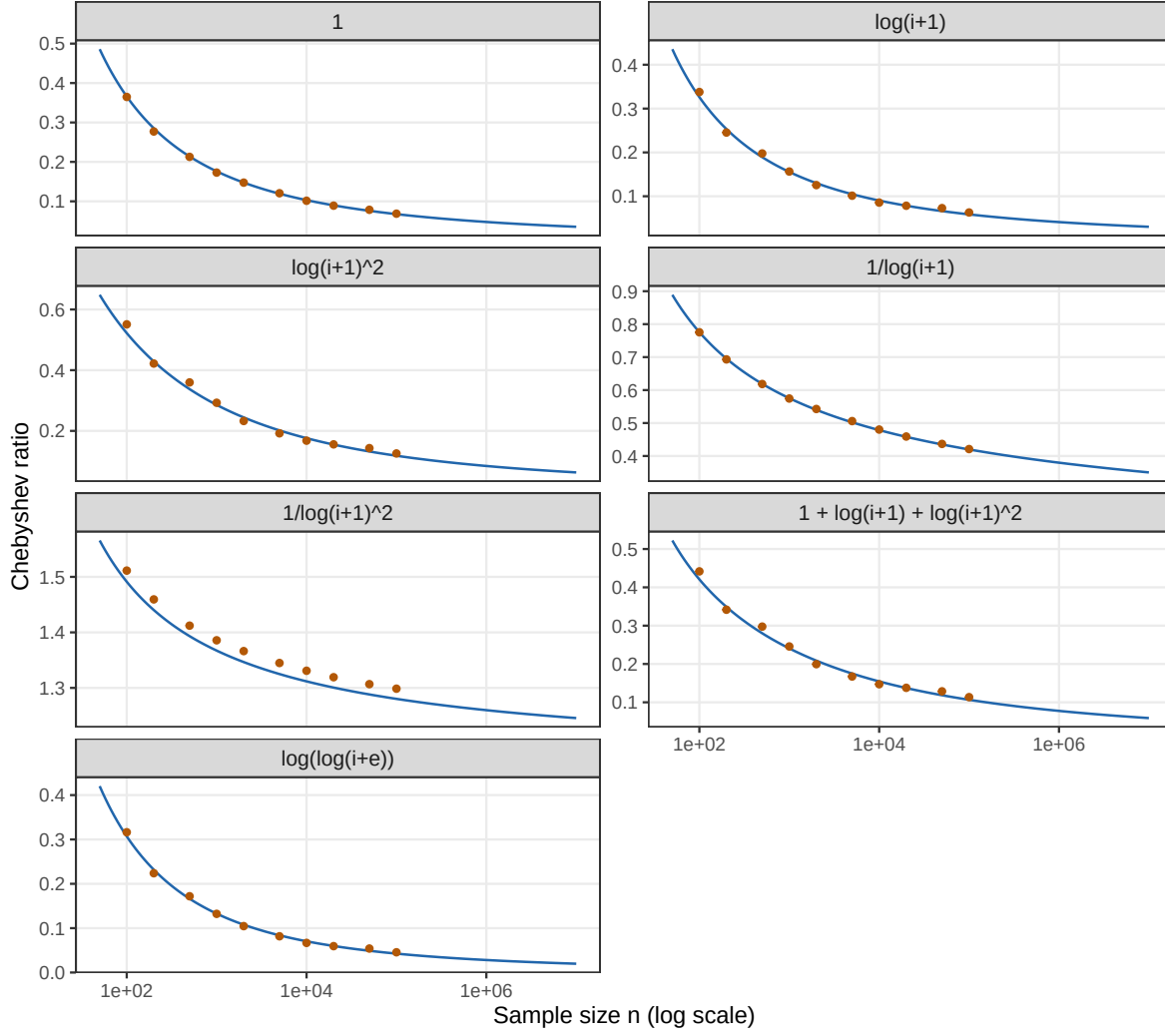


Figure 1: Exact and empirical Chebyshev ratios for the seven main weights.

Do wrong-selection frequencies decrease with sample size?

Points are simulated wrong-selection frequencies and bars are pointwise 95% Wilson intervals. The Monte Carlo standard error is $\{\hat{q}(1-\hat{q})/5000\}^{1/2}$. A zero count still has a positive Wilson upper endpoint.

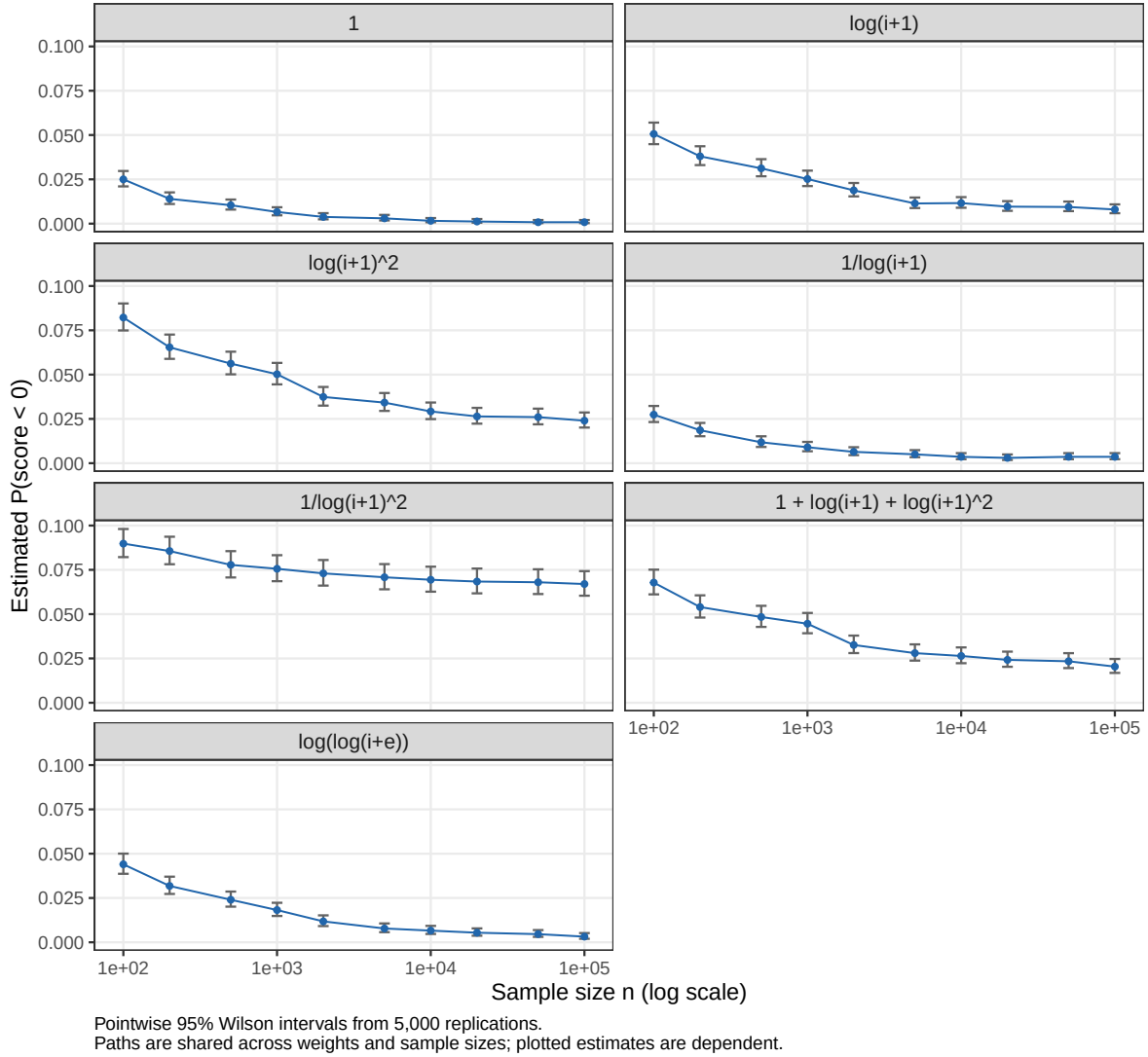


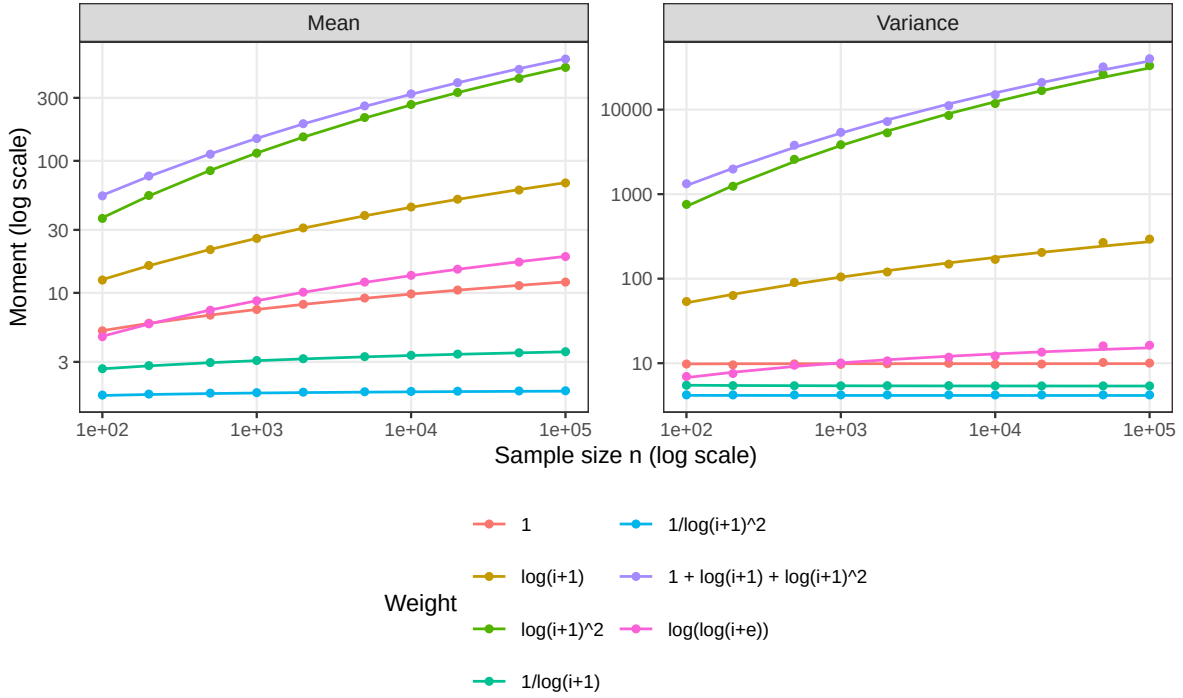
Figure 2: Incorrect-selection frequencies and Monte Carlo uncertainty.

Why is the ratio shrinking?

Figure 3 separates mean and variance. For positive log powers both grow, while for negative log powers the variance has a finite limit.

Table 2: Exact moments and ratios, with error estimates, at $n = 100,000$. Lower and Upper are Wilson endpoints.

Weight	Mean	Variance	Ratio	Error	MCSE	Lower	Upper
1	12.0901	9.8695	0.0675	0.0008	0.0004	0.0003	0.0021
$\log(i+1)$	68.3192	273.7620	0.0587	0.0080	0.0013	0.0059	0.0109
$\log(i+1)^2$	513.4996	31143.6081	0.1181	0.0240	0.0022	0.0201	0.0286
$1/\log(i+1)$	3.5762	5.3690	0.4198	0.0036	0.0008	0.0023	0.0057
$1/\log(i+1)^2$	1.8011	4.1533	1.2803	0.0670	0.0035	0.0604	0.0743
$1 + \log(i+1) + \log(i+1)^2$	593.9089	37596.7760	0.1066	0.0204	0.0020	0.0168	0.0247
$\log(\log(i+e))$	18.8973	15.2116	0.0426	0.0032	0.0008	0.0020	0.0052

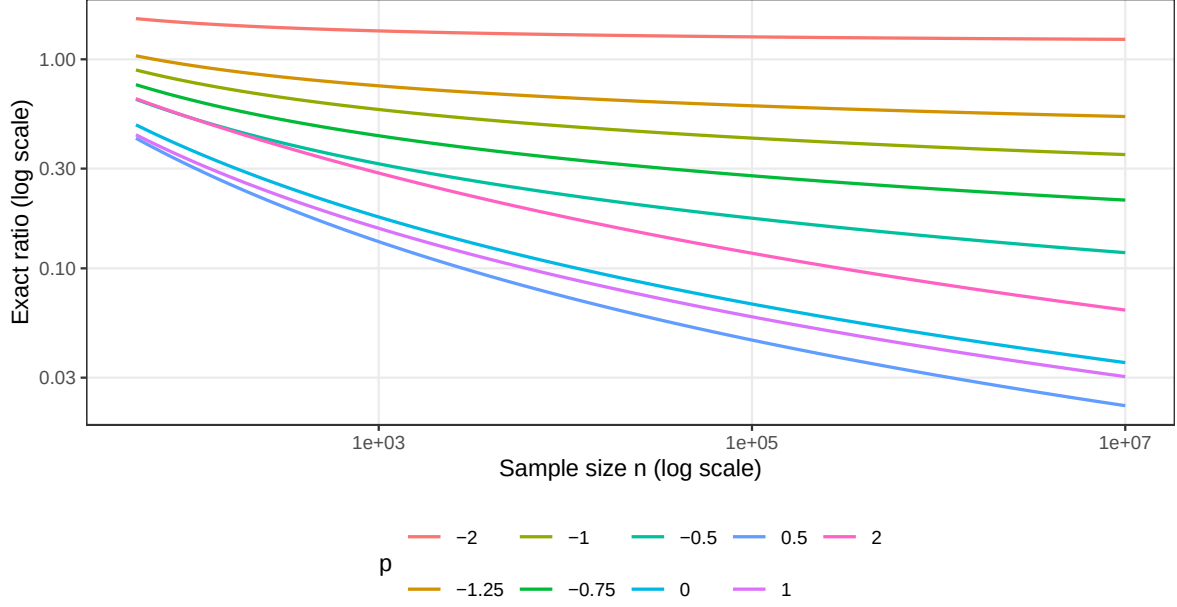


Lines: exact moments. Points: Monte Carlo moments.

Figure 3: Theoretical and empirical means and variances.

Where is the boundary in the log-power family?

Figure 4 shows the $p = -1$ boundary. Its ratio decreases only as $(\log \log n)^{-2}$, explaining the slow finite-sample change.



The ratio tends to zero for $p \geq -1$; it has a positive limit for $p < -1$.
 Slow finite-sample change near $p = -1$ does not establish a plateau.

Figure 4: Exact ratio trajectories within the log-power family.

For $p < -1$, stopping the exact calculation at a finite N leaves out a mean tail that shrinks slowly. The omitted part is asymptotic to $(\log N)^{p+1}/(-p-1)$, even for very large N . Table 3 adds approximate tail corrections and shows how the estimates change as N grows. It uses

$$\widehat{M}_\infty = M_N + \frac{(\log N)^{p+1}}{-p-1}, \quad \widehat{V}_\infty = V_N - 2(\log N)^{2p}.$$

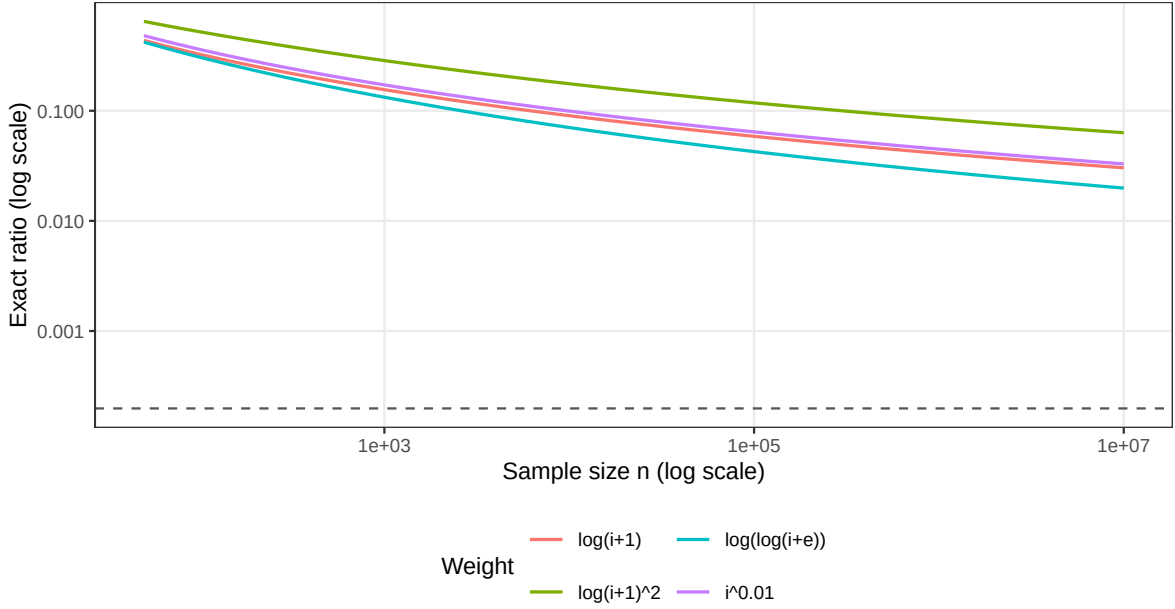
The mean correction has error $O((\log N)^p/N)$ by integral comparison. For the variance, integration by parts gives $W_{n-1} = n(\log n)^p \{1 - p/\log n + O((\log n)^{-2})\}$. Substitution into (8) and summation of the tail give $V_\infty - V_N = -2(\log N)^{2p} + O((\log N)^{2p-1})$. The comparisons show numerical stability across these truncations; the approximations are not certified error bounds.

Table 3: Truncation stability of tail-corrected positive-ratio limits.

p	N	$\widehat{M}_\infty$	$\widehat{V}_\infty$	$\widehat{R}_\infty$
-1.25	1e+05	5.058954	4.993334	0.195105
-1.25	1e+06	5.058954	4.992914	0.195089
-1.25	1e+07	5.058954	4.992733	0.195082
-2.00	1e+05	1.888002	4.153206	1.165141
-2.00	1e+06	1.888002	4.153183	1.165135
-2.00	1e+07	1.888002	4.153175	1.165133

How do these weights compare with a small positive power?

Figure 5 compares three slowly growing weights with $i^{0.01}$ from Analysis 06. Its ratio has the positive limit $2(0.01)^2/1.01 \approx 0.000198$. The dashed limiting value lies far below the power-weight curve at the sample sizes shown. The logarithmic ratios eventually fall below that positive limit because they go to zero, but the figure does not show where the curves cross. A positive ratio limit for the power weight still says nothing decisive about its limiting wrong-selection probability.



Dashed: the positive-power limit $2(0.01)^2 / 1.01$, not a finite- n approximation.
The log-based ratios tend to zero, despite being larger over the displayed range.

Figure 5: Slowly varying weights compared with the small positive power from Analysis 06.

Conclusions by weight family

Baseline: $w_i = 1$

For the unweighted score,

$$M_n \sim \log n, \quad V_n \rightarrow \pi^2, \quad R_n \sim \frac{\pi^2}{(\log n)^2} \rightarrow 0.$$

The unweighted baseline is consistent. The mean signal continues to grow while the variance stays bounded. In the log-power family, $p = 0$ is a special variance regime.

Increasing log-power weights: $p > 0$

For $w_i = [\log(i + 1)]^p$ with $p > 0$,

$$M_n \sim \frac{(\log n)^{p+1}}{p+1}, \quad V_n \sim 2(\log n)^{2p}, \quad R_n \sim \frac{2(p+1)^2}{(\log n)^2} \rightarrow 0.$$

The tested cases are

$$p = 1 : \quad w_i = \log(i + 1), \quad R_n \sim \frac{8}{(\log n)^2},$$

and

$$p = 2 : \quad w_i = [\log(i + 1)]^2, \quad R_n \sim \frac{18}{(\log n)^2}.$$

Every fixed positive log power tested here is consistent. On the increasing side of this family, no Chebyshev-ratio consistency boundary appears.

Rule: Both mean and variance grow, but the squared mean grows two powers of $\log n$ faster than the variance.

Decreasing log-power weights: $p < 0$

Slowly decreasing weights: $-1 < p < 0$

Here

$$M_n \sim \frac{(\log n)^{p+1}}{p+1} \rightarrow \infty, \quad V_n \rightarrow V_\infty(p) < \infty, \quad R_n \rightarrow 0.$$

Slowly decreasing weights are still consistent, so increasing weights are not necessary. One example is

$$p = -\frac{1}{2} : \quad w_i = \frac{1}{\sqrt{\log(i+1)}}.$$

Boundary weight: $p = -1$

For the tested weight $w_i = 1/\log(i+1)$,

$$M_n \sim \log \log n \rightarrow \infty, \quad V_n \rightarrow V_\infty(-1), \quad R_n \asymp \frac{1}{(\log \log n)^2} \rightarrow 0.$$

Reciprocal log is consistent, although its ratio goes to zero very slowly.

Faster decreasing weights: $p < -1$

In this range,

$$M_n \rightarrow M_\infty(p) > 0, \quad V_n \rightarrow V_\infty(p) > 0, \quad R_n \rightarrow \frac{V_\infty(p)}{M_\infty(p)^2} > 0.$$

The tested example is

$$p = -2 : \quad w_i = \frac{1}{[\log(i+1)]^2}.$$

The total mean signal stops growing, so the Chebyshev ratio no longer vanishes. This does not prove inconsistency; it only means that the Chebyshev bound no longer proves consistency.

Therefore,

$$p = -1 \text{ is the Chebyshev-ratio boundary for } [\log(i+1)]^p.$$

Rule: Decreasing weights can still work if they decrease slowly enough. The boundary occurs when the mean changes from diverging to converging.

Polynomial of log

The tested polynomial is

$$w_i = 1 + \log(i+1) + [\log(i+1)]^2.$$

More generally, if

$$P(x) = cx^k + \text{lower-order terms}, \quad c > 0, \quad k \geq 1,$$

then $P(\log(i+1)) \sim c[\log(i+1)]^k$. Consequently,

$$M_n \sim \frac{c}{k+1} (\log n)^{k+1}, \quad V_n \sim 2c^2 (\log n)^{2k}, \quad R_n \sim \frac{2(k+1)^2}{(\log n)^2} \rightarrow 0.$$

For the tested degree-two polynomial, $k = 2$ and $c = 1$, so $R_n \sim 18/(\log n)^2$. The tested polynomial is consistent.

Rule: A positive polynomial of $\log i$ is controlled asymptotically by its highest-degree term. A degree- k polynomial behaves like the log-power case $p = k$.

Log-log weight

For $w_i = \log \log(i+e)$,

$$M_n \sim (\log n) \log \log n, \quad V_n \sim 2(\log \log n)^2, \quad R_n \sim \frac{2}{(\log n)^2} \rightarrow 0.$$

The log-log weight is consistent even though it grows more slowly than $\log i$.

Rule: Very slow increasing growth can still be enough; strongly increasing weights are not necessary.

Overall rules from Analysis 07

1. For $w_i = [\log(i + 1)]^p$, the Chebyshev-ratio boundary is $p = -1$.
2. Every fixed $p > 0$ considered here gives consistency.
3. The range $-1 \leq p < 0$ is also consistent, so weights need not increase.
4. For $p < -1$, both mean and variance converge to positive constants and the ratio has a positive limit. The Chebyshev argument is then inconclusive.
5. A polynomial $P(\log(i + 1))$ is controlled by its highest-degree term: $P(\log(i + 1)) \sim c[\log(i + 1)]^k$. Thus degree k behaves like $p = k$.
6. The log-log weight is consistent, with $R_n \sim 2/(\log n)^2 \rightarrow 0$.

The main lesson is that consistency is not determined simply by whether the weights increase or decrease. In this Chebyshev argument, what matters is whether the accumulated mean signal becomes large enough relative to the score's standard deviation.